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Wednesday, March 11, 2026 15:24

- Programming Lang: **Rust**
  - **Impact**: The biggest impact Rust brings to programming is the introduction of memory safety without the use of a garbage collector
    - Garbage collector: automatic memory management that identifies and reclaims memory no longer being used by the program
      - Used in Java, python, C#, Go, JS, Ruby
      - Avoids manual memory management done in C and C++
      - No longer "trust the programmer"
    - **Solutions**: Ownership system and borrow checker
      - Compile-time analysis enforcing that every value has exactly 1 owner, borrowing is tracked, and data races are impossible by construction.
      - Wont compile if it is not memory-safe
  - **Type of language**: systems programming language mainly from the Algol 60 family but also appears in the Lisp family through influences. Overall is a multi-paradigm meaning it supports Imperative, Functional and object-oriented.
  - **Concept application:**
    - Implementation method: Compilation
      - Translated entirely to machine language before execution starts
    - Language evaluation criteria
      - Safety & reliability: Unmatched
      - Performance: Comparable to C/C++
      - Maintainability and Readability: High BUT has a steep learning curve
        - Code is more maintainable due to explicitness and strong type guarantees
      - Writability: Has a steep learning curve
- Code: https://github.com/mxneo00/Rust Guessing-game